

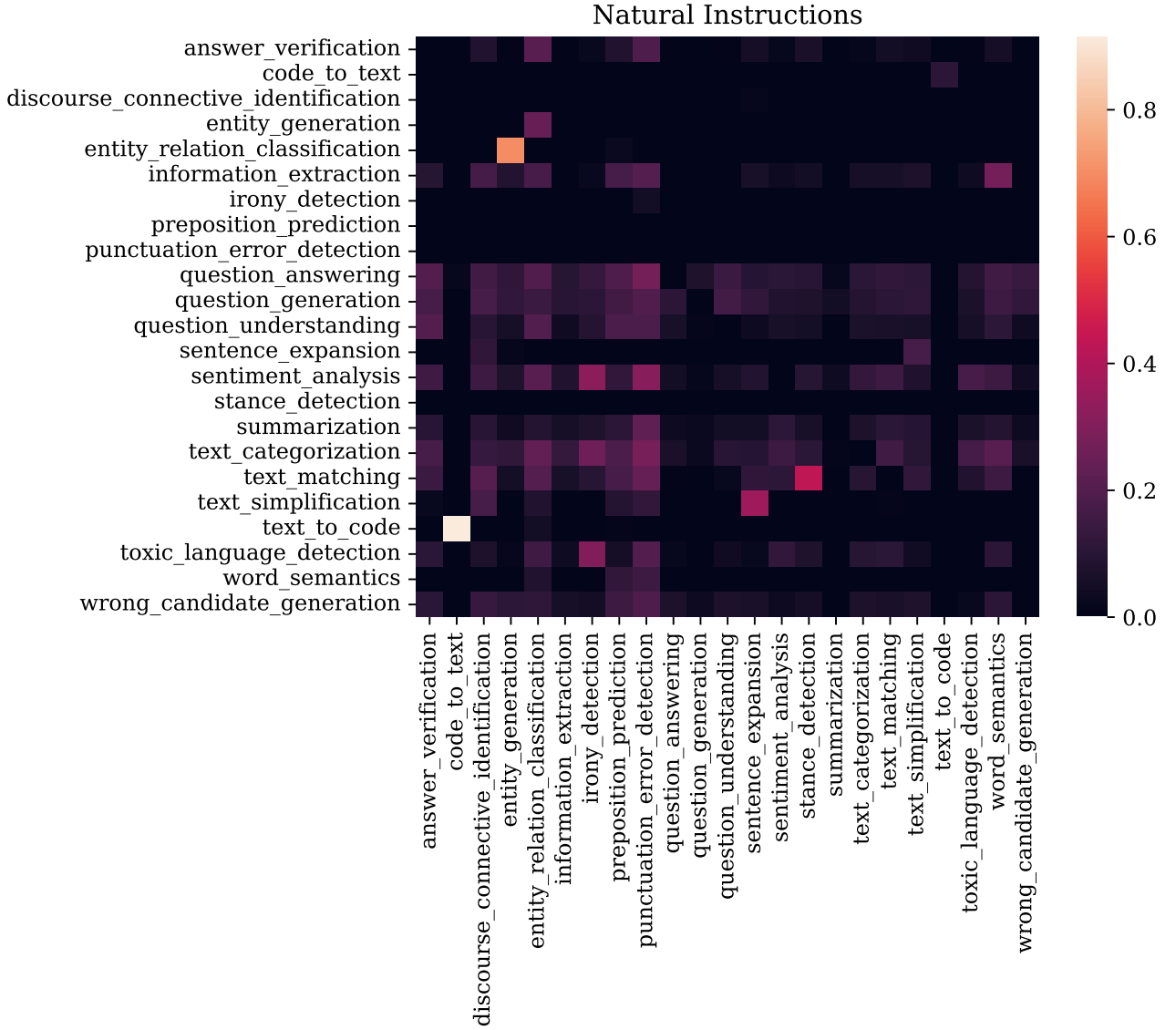


Figure 12: Natural Instructions heatmap where i, j th entry is $\max(0, \delta_j^i)$ (the change in loss on s_j after training on s_i for 100 steps). Diagonal entries are set to 0 for clearer visualization.

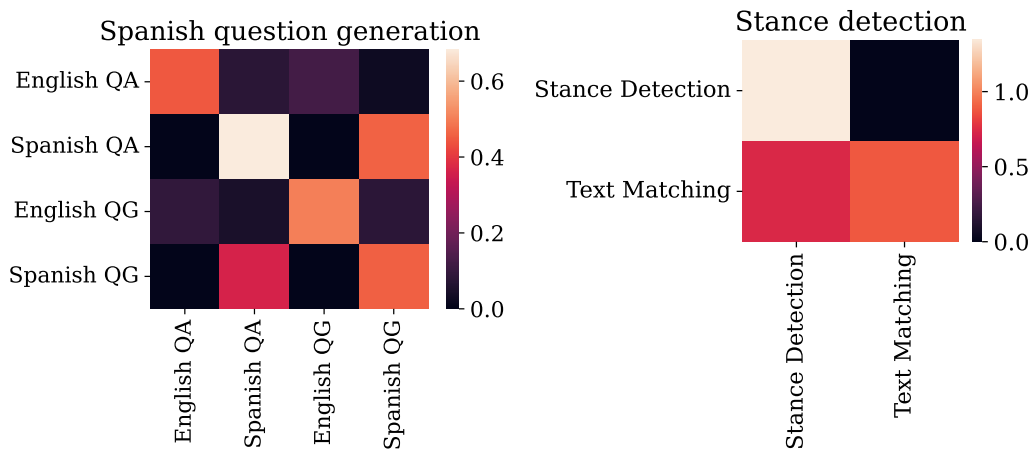


Figure 13: Spanish question generation and stance detection heatmaps where i, j th entry is $\max(0, \delta_j^i)$ (the change in loss on s_j after training on s_i for 100 steps).

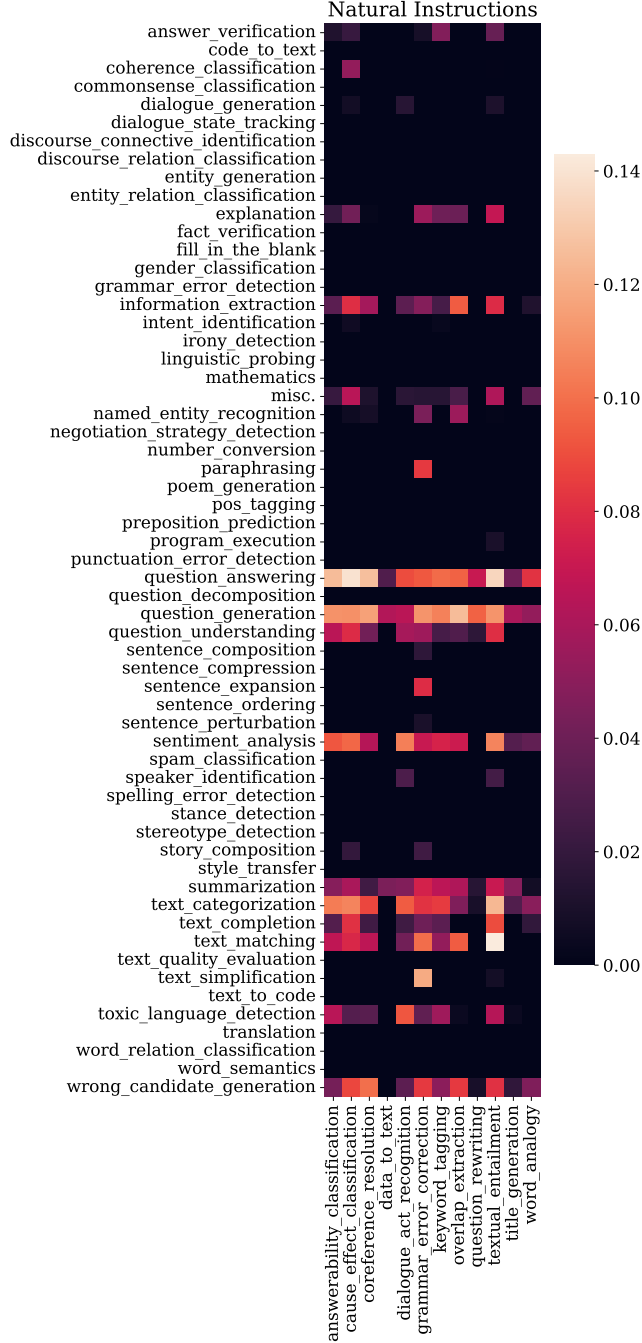


Figure 14: Natural Instructions heatmap for out-of-domain setting where rows are for the training skills and columns are for the evaluation skills. The i, j th entry is $\max(0, \delta_j^i)$ (the change in loss on s_j after training on s_i for 100 steps).

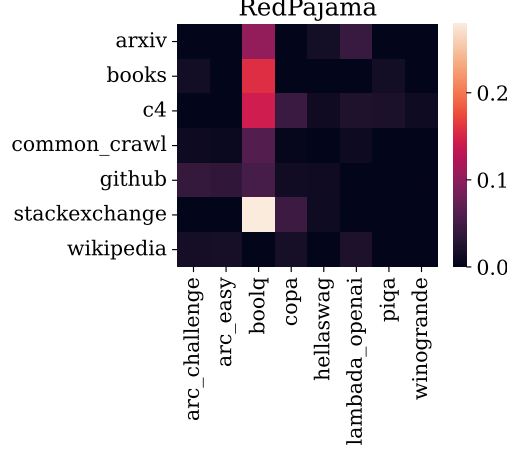


Figure 15: RedPajama heatmap for out-of-domain setting where rows are for the training skills and columns are for the evaluation skills. The i, j th entry is $\max(0, \delta_j^i)$ (the change in perplexity on s_j after training on s_i for 1B tokens).

- LEGO: $\eta = 0.5, T = 6, w = 3$. We train for 6000 steps.
- Addition: $\eta = 0.1, T = 5, w = 3$. We train for 6000 steps.
- Natural Instructions (pre-training): $\eta = 0.2, T = 1$. We train for 5000 steps.

For the LEGO random baseline, when we selected points at random, we used an imbalanced training dataset with proportions 1:1:1:3:5. For the addition random baseline, we used an imbalanced dataset with randomly selected proportions: 13:14:18. For the curriculum learning baselines, the pacing function, $g(i)$, denotes the size of the subset of the highest scoring samples that we uniformly select from in the i th epoch. We define our pacing function as $g(i) = \frac{iH}{M}$, where H is the number of steps and M is 5 epochs for LEGO and NI, and 3 for addition.

SKILL-IT fine-tuning

- LEGO: $\eta = 0.5, T = 10, w = 3$. We train for 6000 steps.
- Addition: $\eta = 0.1, T = 5, w = 3$. We train for 6000 steps.
- Natural Instructions (Spanish QG): $\eta = 0.8, T = 6, w = 3$. We train for 600 steps.
- Natural Instructions (stance detection): $\eta = 0.2, T = 6, w = 3$. We train for 600 steps.

SKILL-IT out-of-domain

- Natural Instructions: $\eta = 0.2, T = 10, w = 3$. We train for 5000 steps.
- RedPajama: $\eta = 100, T = 1$. We train for 3 billion tokens.

All results are computed over 5 random seeds.

Batch sizes of 32 and 64 were used for the LEGO and addition synthetic on the 125M and 1.3B parameter model, respectively. Batch sizes of 4 and 16 were used for the Natural Instructions experiments on the 125M and 1.3B parameter model.

For the out-of-domain Natural Instructions experiment and Alpaca graph learning experiments, a learning rate of $5e-6$ with linear scheduler and 50 warmup steps was used. For the Natural Instructions continual pre-training experiment on the 1.3B parameter model, a learning rate of $1e-6$ was used. All other experiments used a learning rate of $5e-5$. All experiments used AdamW with betas = 0.9, 0.999, eps = $1e-8$, and weight decay = 0.01. A context window of 512 was used for all experiments except LEGO and addition, which used a window of 128.

Experiments with the Addition dataset were run using an Nvidia RTX A6000. Other experiments using the GPT-Neo 125M parameter model were run on an Nvidia Tesla P100. Experiments using the GPT-Neo 1.3B parameter model were run on an Nvidia Tesla A100.